Problem Statement

Factories now let AI inspect every product for defects — but if something later goes wrong, there's no way to prove what the AI actually saw, decided, or whether that record was quietly changed afterward. We've seen this exact failure before: manufacturers editing inspection data to hide problems. Our platform locks the AI's evidence, decision, and any human override together the moment it happens — so nobody, not even us, can rewrite what really occurred.

Case Studies which explains the Problem:

1. Kobe Steel (2017–2018) — Falsified inspection data across 23 domestic and overseas plants, reportedly since the 1970s, with 40+ employees involved. Shipped 20,000+ tons of aluminum/copper with fabricated inspection data to ~200 companies, including automakers and aircraft/bullet-train manufacturers — even triggering U.S. NRC review of nuclear supply chains. **The single biggest real-world proof your problem is real and industry-wide, not hypothetical.**

2. Mitsubishi Materials — Admitted misconduct explicitly described as "rewriting inspection records data." **Closest documented match to your exact attack scenario** — someone with database access simply editing a record after the fact.

3. Global Calcium (FDA, pharma, 2024) — FDA found fabricated batch records for activities that never occurred, directed by plant management. Notably, the files documenting the plan to fabricate records were deleted from a manager's computer the night before an inspection. **Your strongest example of deliberate evidence deletion** — exactly the "silent deletion" attack type your system targets.

4. Nidec (2026) — Recent quality fraud allegations involving 1,000+ cases of unauthorized design changes. **Shows this is an ongoing pattern, not a closed historical chapter** — useful for countering "hasn't this already been solved" pushback.

The Solution, in Stages

Stage 1 — Detect An AI camera system inspects every product as it comes off the line, spotting defects — cracks, dents, bad welds — and recording its decision along with how confident it is..

Stage 2 — Bind The moment a decision is made, we lock together everything that matters: the exact image, which AI version made the call, its reasoning, and the confidence score — as one sealed unit.

Stage 3 — Anchor That sealed unit gets a unique digital fingerprint, recorded on a public, tamper-evident ledger. Routine checks are grouped together to save cost; serious defects get recorded immediately, without delay.

Stage 4 — Respond If a human operator disagrees with the AI and overrides it, that decision — who, why, and what evidence they reviewed (employee metadata)— gets sealed and recorded the same way, permanently.

Stage 5 — Verify Anytime later, anyone can re-check a record against its original fingerprint. If even one detail was altered, the mismatch is caught and flagged immediately — no exceptions, no cover-ups.

Stage 6 — Report All verified records automatically compile into audit-ready reports for inspectors, regulators, or customers — proof the factory's quality process actually happened the way it claims it did.

STAKEHOLDERS

Primary stakeholders **(inspection team is not day today user)**

- **Quality Assurance / Inspection team** — the actual day-to-day users; review AI decisions, override when needed, investigate defects.
- **Plant / Manufacturing management** — monitor overall defect trends and inspection effectiveness across lines.
- **Wrongly-flagged or disputed cases** — the party affected when a record is contested; needs a fair resolution path, easy to forget as a stakeholder but genuinely one.

Secondary stakeholders

- **Compliance & audit teams** — independently verify inspection records during internal or external reviews.
- **External regulators** — validate manufacturing history without relying solely on the company's own internal claims.
- **Customers / OEM buyers** — for high-value components, may want proof a batch actually passed inspection before accepting it.
- **Insurers** — during an incident investigation, need to confirm required inspections genuinely occurred.
- **AI/data science team** — tracks model versions, performance drift, and override patterns over time.

CLOSE ENOUGH EXISITING SOLUTION

Cognex makes AI cameras that spot defects on factory production lines — a real, widely-used product. Their latest update added "Audit Logging," meaning they now keep a record of inspections too. But it's unclear if that record is tamper-proof — nothing shows it's locked against being quietly edited later.